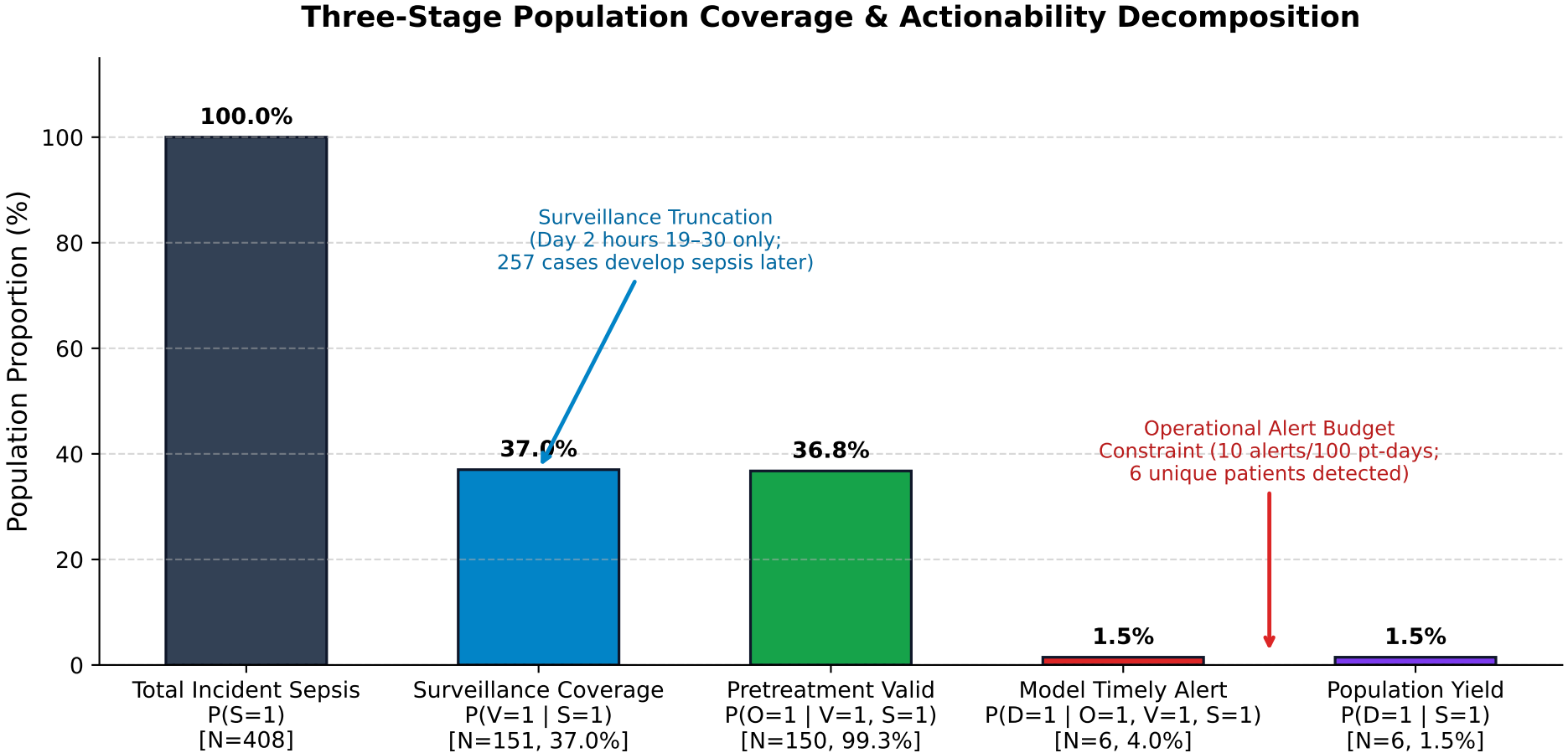
Three-Stage Population Coverage & Actionability Decomposition
Population Proportion (%)
0
20
40
60
80
100
100.0%
37.0%
36.8%
1.5%
1.5%
Surveillance Truncation
(Day 2 hours 19–30 only;
257 cases develop sepsis later)
Operational Alert Budget
Constraint (10 alerts/100 pt-days;
6 unique patients detected)
Total Incident Sepsis
P(S=1)
[N=408]
Surveillance Coverage
P(V=1 | S=1)
[N=151, 37.0%]
Pretreatment Valid
P(O=1 | V=1, S=1)
[N=150, 99.3%]
Model Timely Alert
P(D=1 | O=1, V=1, S=1)
[N=6, 4.0%]
Population Yield
P(D=1 | S=1)
[N=6, 1.5%]